BUSINESS UNDERSTANDING REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 1 Assignment

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1. Introduction

Machine Learning has become an essential technology for helping organizations make data-driven decisions. Businesses across industries use Machine Learning to identify patterns in historical data, automate decision-making, and predict future outcomes. In the telecommunications industry, one of the most important applications of Machine Learning is customer churn prediction.

Customer churn occurs when customers discontinue their relationship with a company by cancelling or failing to renew their services. High customer churn negatively affects business revenue, profitability, and market competitiveness. Retaining existing customers is generally more cost-effective than acquiring new ones, making customer retention a strategic priority for telecommunication companies.

This project focuses on developing a Machine Learning solution for ABC Communications Ltd to predict customer churn using historical customer data. The insights generated from the model will enable the company to identify customers who are at risk of leaving and implement proactive retention strategies before churn occurs.

2. Company Background

ABC Communications Ltd is a telecommunications company that provides communication and internet services to residential and business customers. The company offers products such as phone services, internet connectivity, streaming services, online security packages, technical support, and various subscription plans.

Like many organizations operating within the telecommunications industry, ABC Communications Ltd faces increasing competition from other service providers. Customers can easily switch providers whenever they are dissatisfied with service quality, pricing, customer support, or available service packages.

As customer turnover increases, the company's operational costs also increase because attracting new customers requires significant investment in advertising, promotions, and onboarding activities. Consequently, improving customer retention has become one of the company's primary business objectives.

To address this challenge, ABC Communications Ltd intends to adopt Machine Learning techniques that can accurately predict customer churn and support proactive customer retention initiatives.

3. Business Problem

ABC Communications Ltd currently lacks an automated system capable of identifying customers who are likely to discontinue their services.

Without accurate predictions, the company reacts only after customers have already cancelled their subscriptions, resulting in lost revenue and missed opportunities for customer retention.

The business therefore requires a predictive Machine Learning model capable of analysing customer information and identifying customers who exhibit characteristics similar to those who previously churned.

Such a model would enable management to implement targeted retention strategies such as promotional offers, personalized discounts, loyalty rewards, improved customer support, and service upgrades before customers decide to leave.

The primary business problem addressed in this project is therefore the prediction of customer churn to improve customer retention and reduce revenue loss.

4. Machine Learning Overview

Machine Learning is a branch of Artificial Intelligence that enables computer systems to learn patterns from historical data and make predictions without being explicitly programmed for every decision.

Rather than relying on fixed rules, Machine Learning algorithms discover relationships between customer characteristics and historical outcomes. These learned relationships are then applied to new customer records to predict future behaviour.

In this project, Machine Learning will analyse customer demographic information, account details, subscribed services, billing history, and contract information to determine whether a customer is likely to churn.

The project represents a supervised learning problem because historical customer records include known outcomes indicating whether each customer stayed with or left the company. These labelled examples enable the model to learn the characteristics associated with customer churn.

5. Predictive Analytics

Predictive Analytics is the process of using historical data, statistical techniques, and Machine Learning algorithms to estimate future outcomes.

Instead of describing what has already happened, predictive analytics attempts to answer questions about what is likely to happen next.

For ABC Communications Ltd, predictive analytics will be used to estimate the probability that an existing customer will discontinue the company's services.

The prediction process involves analysing customer behaviour, service usage, billing information, contract type, and demographic characteristics to identify patterns associated with previous customer departures.

The resulting predictions allow business managers to make proactive decisions instead of reactive ones, thereby improving customer retention and overall business performance.

6. Understanding Customer Churn

Customer churn refers to the loss of customers who discontinue using a company's products or services during a specified period.

Within the telecommunications industry, churn may occur for several reasons, including:

- Poor customer service.
- High subscription costs.
- Attractive offers from competitors.
- Unsatisfactory internet or network performance.
- Expiration of customer contracts.
- Dissatisfaction with available service packages.

High customer churn negatively affects organizational growth because existing customers generate recurring revenue over time. Losing loyal customers reduces profitability while increasing customer acquisition costs.

Predicting churn before customers leave enables organizations to implement customer retention strategies that improve long-term profitability and strengthen customer relationships.

7. Business Questions and Answers

Question 1: What business problem should the model solve?

The Machine Learning model should predict which customers are likely to discontinue their services. By identifying high-risk customers before they leave, ABC Communications Ltd can implement targeted retention strategies, reduce customer attrition, and protect future revenue.

Question 2: What is the target variable?

The target variable is **Churn**.

This variable indicates whether a customer has discontinued the company's services.

Possible values include:

- Yes – Customer churned.
- No – Customer remained with the company.

Since the model predicts one of two possible outcomes, the target variable represents a binary classification problem.

Question 3: Which input features should be used?

The model should use customer attributes that may influence churn, including:

- Gender
- Senior Citizen status
- Partner
- Dependents
- Tenure

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Technical Support
- Streaming TV
- Streaming Movies
- Contract Type
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges

The **customerID** field should be excluded because it uniquely identifies customers but does not contribute meaningful predictive information.

Question 4: Is this a classification or regression problem?

This project is a **binary classification** problem.

The objective is to predict whether a customer belongs to one of two classes:

- Churn = Yes
- Churn = No

Regression techniques are not appropriate because the outcome is categorical rather than continuous.

Question 5: Which evaluation metrics should be used?

The following evaluation metrics are appropriate for this project:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix

Although accuracy provides an overall measure of model performance, Recall and F1-Score are particularly important because failing to identify customers who are likely to churn could result in significant business losses.

Question 6: Which Machine Learning algorithms are suitable?

Several supervised learning algorithms are appropriate for customer churn prediction, including:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- XGBoost
- LightGBM

Among these, Random Forest and XGBoost are widely used in customer churn prediction because they can effectively model complex relationships within customer behaviour data while providing high predictive performance.

8. Project Objectives

The primary objective of this project is to develop a Machine Learning solution capable of predicting customer churn using historical customer information.

Specific objectives include:

- Understand the business problem from a Machine Learning perspective.
- Analyse the customer churn dataset.
- Identify important customer characteristics influencing churn.
- Prepare the dataset for Machine Learning.
- Recommend appropriate Machine Learning algorithms.
- Support proactive customer retention strategies.
- Reduce customer attrition.
- Improve organizational decision-making.

9. Expected Business Benefits

Successful implementation of the proposed Machine Learning solution will provide several business benefits for ABC Communications Ltd.

These include:

- Early identification of customers likely to churn.
- Improved customer retention through proactive intervention.
- Reduced revenue loss caused by customer departures.
- More efficient allocation of marketing and customer support resources.
- Improved customer satisfaction and loyalty.
- Better strategic decision-making through data-driven insights.
- Increased long-term profitability.
- Stronger competitive advantage within the telecommunications industry.

By acting on model predictions before customers leave, the company can significantly reduce churn and strengthen long-term customer relationships.

10. Conclusion

Customer churn prediction represents one of the most valuable applications of Machine Learning within the telecommunications industry. By leveraging historical customer data, ABC Communications Ltd can accurately identify customers who are likely to discontinue their services and implement timely interventions to improve customer retention.

This report has presented the business context, defined the customer churn problem, explained the role of Machine Learning and predictive analytics, answered the project's business questions, and outlined the expected benefits of implementing a predictive churn model.

The next phase of the project will focus on inspecting the dataset, evaluating data quality, identifying preprocessing requirements, and performing exploratory data analysis to prepare the data for model development.